

CENTRALIA, IL, USA

WMO: 744657

Lat: 38.515N

Lon: 89.092W

Elev: 163

StdP: 99.38

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53887

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.7	-11.1	-19.6	0.7	-12.4	-17.1	0.9	-10.4	12.1	5.5	11.0	4.3	3.0	330	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	34.8	24.7	33.7	24.6	32.2	24.1	27.1	32.2	26.2	31.2	25.3	30.0	3.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.8	21.6	30.4	24.8	20.3	29.4	23.8	19.1	28.2	86.5	32.7	82.0	31.3	78.1	30.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.7	7.7	-18.0	37.0	2.7	1.7	-19.9	38.2	-21.5	39.2	-23.0	40.1	-24.9	41.3		
(5)			-18.5	28.2	2.6	1.7	-20.4	29.4	-21.9	30.4	-23.4	31.3	-25.3	32.6	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	-0.2	1.4	7.8	13.7	19.1	24.0	25.3	24.5	20.7	14.2	7.5	2.4	(6)
(7)		DBStd	10.31	6.36	6.49	6.04	5.12	4.67	3.21	2.98	3.14	3.90	5.33	5.41	5.64	(7)
(8)		HDD10.0	1078	319	250	116	22	1	0	0	0	0	18	110	243	(8)
(9)		HDD18.3	2577	573	474	329	153	50	2	0	2	20	152	327	494	(9)
(10)		CDD10.0	2330	4	9	47	132	284	418	476	449	321	148	34	7	(10)
(11)		CDD18.3	788	0	0	2	14	74	171	218	193	91	24	1	0	(11)
(12)		CDH23.3	7719	0	0	12	114	616	1659	2317	1898	881	215	8	0	(12)
(13)		CDH26.7	3052	0	0	1	11	174	663	1022	803	321	57	1	0	(13)
(14)	Wind	WSAvg	3.1	4.1	3.8	4.0	3.9	3.1	2.6	2.0	1.8	1.9	2.8	3.4	3.6	(14)
(15)	Precipitation	PrecAvg	1064	82	59	82	101	137	109	95	77	77	80	100	68	(15)
(16)		PrecMax	1428	192	116	174	253	360	250	215	167	204	157	184	211	(16)
(17)		PrecMin	839	12	15	21	32	26	20	51	23	7	29	15	26	(17)
(18)		PrecStd	150	47	30	37	58	85	56	45	36	48	39	47	41	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	20.7	25.5	28.0	32.5	35.2	37.9	37.2	34.3	31.2	24.0	18.8	(19)
(20)			MCWB	15.7	16.6	17.6	19.6	23.2	23.4	23.7	24.9	22.6	22.4	17.2	15.8	(20)
(21)		2%	DB	15.4	17.5	22.5	26.2	30.2	33.8	34.8	34.7	32.5	27.9	20.9	15.9	(21)
(22)			MCWB	13.3	14.0	16.3	18.7	22.3	24.7	25.4	24.7	23.4	19.9	16.5	13.6	(22)
(23)		5%	DB	12.7	14.4	19.9	24.7	28.7	32.5	33.7	32.9	30.3	25.8	18.2	13.5	(23)
(24)			MCWB	10.3	11.0	15.0	17.9	21.6	23.8	25.2	24.3	22.8	19.0	14.3	11.0	(24)
(25)		10%	DB	9.4	11.3	17.5	22.6	27.1	30.3	31.8	31.0	28.0	23.1	16.3	11.0	(25)
(26)			MCWB	7.1	8.6	13.0	16.7	20.6	22.9	24.5	23.7	21.2	17.7	12.2	8.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.5	17.3	19.0	21.5	25.1	27.3	29.8	28.5	26.0	23.6	19.8	16.5	(27)
(28)			MDCB	17.8	19.9	23.6	26.4	29.9	33.5	34.3	33.5	31.1	28.3	22.9	18.3	(28)
(29)		2%	WB	13.8	14.9	17.0	20.0	23.7	26.1	27.4	26.9	25.0	22.1	17.1	13.9	(29)
(30)			MDCB	15.5	17.0	21.4	25.0	28.4	31.9	32.0	31.9	30.0	26.4	19.5	15.8	(30)
(31)		5%	WB	10.6	11.6	15.5	18.7	22.6	25.1	26.3	25.8	23.9	20.6	14.8	11.3	(31)
(32)			MDCB	12.4	14.1	19.2	23.1	27.1	30.3	31.1	30.5	28.5	23.9	18.0	13.2	(32)
(33)		10%	WB	6.7	8.5	13.8	17.4	21.5	23.9	25.3	24.8	22.6	18.6	12.6	8.6	(33)
(34)			MDCB	9.3	11.5	16.9	21.9	25.8	28.4	29.9	29.1	26.5	22.3	15.8	10.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.2	8.7	10.0	11.1	10.5	11.0	11.6	12.5	11.9	9.8	7.7	(35)	
(36)			MCDBR	11.5	12.6	12.3	13.1	11.8	12.6	12.6	13.5	14.2	14.0	12.2	10.5	(36)
(37)			MCWBR	9.8	10.2	8.5	8.1	6.3	6.3	6.0	6.5	6.6	7.4	8.6	8.8	(37)
(38)		5% WB	MCDBR	10.8	12.0	11.4	11.6	10.4	10.9	10.5	11.2	11.6	11.5	10.9	9.7	(38)
(39)			MCWBR	10.2	10.4	8.4	7.9	6.4	6.7	6.4	6.6	6.5	7.0	9.0	8.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.322	0.361	0.400	0.435	0.437	0.464	0.462	0.408	0.357	0.318	0.304	(40)	
(41)		taud	2.456	2.405	2.361	2.254	2.213	2.233	2.159	2.154	2.294	2.436	2.489	2.482	(41)	
(42)		Ebn at Noon	870	903	897	878	850	846	818	810	836	852	852	849	(42)	
(43)		Edh at Noon	81	98	113	134	142	139	149	146	120	94	78	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.00	2.76	3.73	4.93	5.53	6.41	6.23	5.64	4.71	3.41	2.30	1.69	(44)	
(45)		RadStd	0.20	0.30	0.36	0.28	0.48	0.42	0.41	0.31	0.32	0.36	0.25	0.17	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+78

Nomenclature: See separate page